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Want to improve your golf game? Work on GIR.

Riccio's Rule: Greens in regulation is the most powerful predictor of score.

By Lucius Riccio

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In a comprehensive statistical study performed in 1986 and reported at the First World Scientific Congress of Golf in 1990 [1], each and every stroke from more than a thousand rounds of golf played by over 100 golfers of all skill levels – from Jack Nicklaus in winning the U.S. Open to golfers who struggle to break 100 on their local course – we analyzed to find the fundamental relationships between all aspects of play and their effect on scoring for each level of golfer. In that study we found that greens in regulation (GIRs) were by far the most powerful predictor of score. This was true for explaining the difference in play between the premier players and higher handicap players or the difference between the best and the worst rounds of individual golfers. [A GIR is recorded when the number of strokes a golfer takes to reach the putting surface is equal to the par of the hole minus two. As such, if a golfer reaches the putting surface in two strokes for a par 4, that player is said to have recorded a GIR.]

Although other factors we found to be significant, none came close to the predictive power of GIRs. That analysis yielded the rule that score equals 95 minus 2 times GIRs. The formula, also known as Riccio's Rule or the "f = m a" of golf, is:

Score = 95 - 2 * GIRs (Eq. 1).

Although it is not perfect for any given round, when four or more rounds are averaged, its accuracy is greatly improved. Its predictive accuracy is approximately equal to plus or minus 1 stroke about 90% of the time when averaged over four rounds.

Its predictive/explanatory utility has been proven in a large number of situations, from Tiger Woods' entire 2000 season average score (within 0.18 strokes of year's average from year's GIRs per round), to Greg Norman's meltdown at the 1996 Masters Tournament (averaged 14 GIRs and 67.7 in the first three rounds and 78 and 8 GIRs in the last round), to *LINKS Magazine* editor George Peper's attempt at qualifying for the U.S. Amateur (two rounds of 81 with 7 GIRs in each). A second set of data from a wide range of golfers was statistically analyzed in the same way as the first with a virtually identical result. The intercept was calculated at 98 and the GIR coefficient at 2.1, a difference that proved statistically insignificant.

The implications of the rule can be stated in a variety of ways. First, if a golfer is to improve his scoring significantly, the rule indicates that a golfer must greatly improve his tee-to-green play. Putting is, of course, important, but even if Ben Crenshaw did the putting for a 95-shooter, he would still rarely break 90. Whereas a 95-shooter playing off a typical pro's tee ball would likely play in the mid-80s [2]. A simple way to remember the rule is "3 greens break 90, 8 greens break 80 and 13 greens break 70."

A second implication is that given a golfer's score and his greens in regulation, a simple assessment of that golfer's play can be made. That is, a golfer who hits five GIRs (prediction: 85) but averages 88 is losing about three strokes due to poor short game play, or has a few "blow up" holes. Conversely, a golfer who averages seven greens but regularly shoots 78, probably has a deft touch around the greens. An example of this was the first-round play of Tom Watson at the 1994 U.S. Open when he hit 12 greens in regulation but putted extremely well and scored a 67. In the press tent he said, "Today I turned a 71 into a 67." By the rule, 12 GIRs would have yielded a 71.

As mentioned above, in the study that produced the rule, a large number of other factors were examined for their predictive capabilities. Everything from the number of fairways hit to the average up and down percentage from greenside bunkers was statistically measured to see how well each related to score. Again, none performed as well

as GIRs, but several did pass a significance test. Putting was one that had enough predictive power to warrant further analysis.

The formula for putting generated by the analysis was:

Score = 3 * (No. of Putts) – 16 (Eq. 2).

Actually, the constant 16 was 15.8, but rounding to 16 is not inappropriate given the amount of error in the estimate.

Using this equation, we can predict the score of someone who takes, say, 30 putts per round. The formula would predict that someone who took 30 putts would score 3*(30)-16, or 90 minus 16, or finally 74. The typical PGA Tour pro scores about 71 to 72 and takes just more than 29 putts on average. So, for a 29-putts-per-round player, the equation would predict 3*(29)-16, or 87 minus 16, or 71. It should be pointed out that there is a tremendous amount of variability to this prediction, about three times more than with Eq. 1. As such, someone with 30 putts could score anywhere between the high 60s and low 80s.

Given this additional equation, I considered combining the two equations to greatly enhance the explanatory or predictive capability of Riccio's Rule. The logic behind this effort was that although the basic rule (Eq. 1) was quite powerful in broad analyses, on any given round it could be off by several strokes, as witnessed above by the Tom Watson experience. Adding in another factor would likely improve the power of the equation.

The basic rule (Eq. 1) "assumes" the golfer will play in all other dimensions of the game along the lines of someone who typically plays in that scoring range. A pro not only plays tee to green like a pro, but also putts and chips like a pro, and a 90-shooter plays tee to green and putts like a 90-shooter. The typical PGA Tour pro hits on average about 12 greens and takes about 29 to 30 putts. Twelve greens would yield 71 from the basic GIR rule and the 30 or so putts just about the same from the putting rule. On the other hand, someone who hits three greens would usually score about 89. That assumes that that golfer putts like an 89-shooter. The putting rule would indicate that that the "89-shooter" would take on average about (89+16)/3 putts, or 105/3, or 35 putts. This seems reasonable. Although there are 89-shooters who can putt like a pro, they are rare.

The two strongest factors linked to scoring were, as mentioned, GIRs and after that putting. I therefore assumed that the majority of any deviation from the prediction from the basic rule could be explained by the difference in the actual number of putts taken versus the predicted or expected number based on the putting rule. That notion would yield an equation that indicated that score was equal to the basic rule prediction from just GIRs plus the difference in actual vs. expected putts. As such, we can say:

Score = 95 – 2 * GIRs + (Actual Putts - Expected Putts Given GIRs) (Eq. 4).

The expected putts would come from the combination of the GIR rule and putting rule. To get expected putts given GIRs, we have to transform Eq.2 as we did above into:

Expected Putts = (Score +16) / 3 (Eq. 5).

Because Eq. 1 relates score to GIRs, we can substitute in the following way:

Expected Putts Given GIRs = (95 – 2 * GIRs + 16) / 3 (Eq. 6).

This yields an expected putts given the golfer's GIRs for the round. This can be simplified as:

Expected Putts Given GIRs = 37 – (2/3) * GIRs (Eq. 7).

From this we can see that a golfer who hits no GIRs is expected to take 37 putts and one who hits 18 GIRs 25 putts. One who hits 12 GIRs should take 29 putts, just about what the average Tour pro does.

Substituting Eq. 7 into Eq. 4, we get the following:

Score = 95 -2 * GIRs + Putts – 37 + (2/3) * GIRs (Eq. 8).

This too can be simplified to:

Score = 58 – (4/3) * GIRs + Putts (Eq. 9).

Equations 1, 7 and 9 constitute three useful, powerful and refined formulas for understanding how scores, GIRs and

taking 35 putts (happens every now and then on the Tour, particularly at venues like the Masters). The maximum likelihood line through the spreadsheet is the path described by Eq. 1, the original Riccio's Rule. There is, however, quite a spread around that line.

The formulas have been tested by numerous golfers with a reasonable degree of success. Most recently one golfer (Tom Perkowski of Indiana) sent me his data for (can you believe!) 2,030 rounds of golf. I fit his data of scores to his GIRs and putts. The regression results are shown in Table 1.

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	58.68438	0.464591	126.3142	0	57.77325	59.5955
GIR	-1.27378	0.016795	-75.8425	0	-1.30672	-1.24084

Putts	0.976575	0.016628	58.72926	0	0.943964	1.009185
<i>Regression Statistics</i>						
Multiple R	0.877945					
R Square	0.770788					
Adjusted R Square	0.770562					
Standard Error	1.822794					
Observations	2030					

Table 1: Regression results based on 2,030 rounds by one golfer.

The intercept of 58.86 and the coefficients for GIRs (-1.274) and putts (0.977) were so close to 58, -4/3 and 1 to confirm the model for this player, anyway.

For the interested golfer, if GIRs are the most important predictor of score, how does a golfer improve their GIRs? In the original study, the factor most highly correlated to GIRs was fairway drives and Par 3 tee iron play accuracy. Work on those shots and your score will improve.

So, in conclusion I believe I have done a pretty thorough study of the matter and come to the belief that Equations 1, 7 and 9 are a viable and powerful set of equations suitable for predictive and explanatory purposes.



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